



Bootstrapping

Megan Ayers
 Stat 141 | Fall 2026
 Monday, Week 7

Goals for Today

- Review how sampling distributions can be used to assess sampling variability
- Discuss bootstrapping as a way of approximating the sampling distribution

Sampling Distribution: Polling Example

An October 2020 poll by Marist College surveyed by phone, asking

If November's election were held today, whom would you support?

- 50% of respondents supported Biden/Harris, 46% supported Trump/Pence, 1% supported another candidate, and 3% were undecided
- In the Nov. 3 2020 election, Biden/Harris had 50.01% of the vote, while Trump/Pence had 48.84% of the vote.
 - **Population:** All registered voters in Pennsylvania ($N \approx 9$ million)
 - **Population Parameter:** The proportion p of registered voters who plan to vote for Trump/Pence. Based on election results, $p = 0.4884$.
 - **Sampling Method:** SRS(?) of size $n = 1020$ obtained using phone-numbers
 - **Sample Statistic:** The sample proportion $\hat{p}$ of Americans who plan to vote for Trump/Pence. In this case, $\hat{p} = 0.46$.

Sampling Variability

How confident should we be in the accuracy of our estimate of $\hat{p} = 0.46$?

- There are about 9 million registered voters in Pennsylvania. Marist College surveyed only 1020 of them (0.01% of the population)
- We should be skeptical that our estimate is *exactly* equal to true proportion.
- But we should feel confident that our estimate is *close* to the true proportion.

Why?

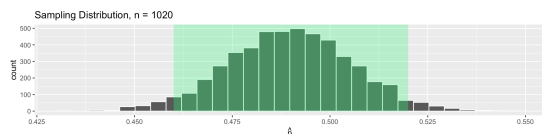
- The sampling distribution tells us how much variability to expect from sample to sample.
- Using probability theory, we know standard error for the sampling distribution for the sample proportion with sample size n is $SE = \sqrt{\frac{p(1-p)}{n}}$
- With $n = 1020$ and $p = 0.4884$, the standard error is $SE \approx 0.016$.

Sampling Variability

Suppose the true proportion of support for Trump/Pence was actually $p = 0.49$

- Reminder: In our sample, $\hat{p} = 0.46$ (0.03 away from $p = 0.49$).

Let's draw 5000 simulated samples of size 1020 to see how many have $\hat{p}$ far from $p = 0.49$.



- In 95% of samples, the sample proportion $\hat{p}$ is at most 0.03 away from the true proportion p !
 - Implication: If 49% of voters support Trump/Pence, then $\approx 95\%$ of samples of size 1020 will show Trump/Pence's support, $\hat{p}$, to be $46\% \leq \hat{p} \leq 52\%$
 - Q: How does this contextualize conclusions based on the poll's sample statistic?

5

The Problem

- For sampling distributions that are $\approx$ bell-shaped, 95% of sample statistics will be within 2 standard errors of the true parameter.
- This helps us assess how close a sample statistic tends to be to the population parameter.
- But in practice, we don't know the sampling distribution!
 - In the polls example, we guessed what the true p was to get a sampling distribution.
 - This is helpful as a thought experiment, but what if our guess was way off?
 - In order to form the actual sampling distribution, we need to collect many, many samples.
 - In the real world, we usually only have one sample!
- The fix?

6

Bootstrapping

Bootstrapping

- The term *bootstrapping* refers to the phrase "to pull oneself up by one's bootstraps"
- Originated in the 19th century as reference to a ludicrous or impossible feat
- By mid 20th century, meaning had changed to suggest a success by one's own efforts, without outside help (the "American Dream" myth)
- Its use in statistics (dating from 1979) alludes to both interpretations.



7

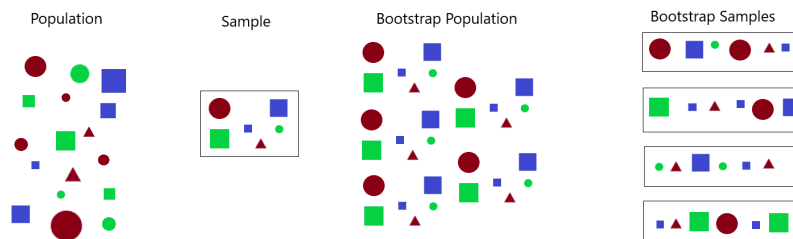
8

The Bootstrap Trick

- The Impossible Task:
 - How can we learn about the sampling distribution, if we only have 1 sample?
- The “Ludicrous” Solution obtained without outside help:
 - Draw repeated samples from the **original sample**
 - Compute the statistic of interest in each new sample, and plot the resulting distribution
- The Main Idea:
 - The original sample approximates the population
 - Resampling from the sample approximates sampling many times from the population
 - The distribution of statistics from the resamples approximates the sampling distribution

9

Theory



- The sample “represents” the population. Many copies of the sample still “represents” the population
- **Idea**: Copy the sample many times to create “a bootstrap population”, and then sample from this to get “bootstrap samples”
- Same result, save time: **Just sample with replacement from the one original sample to get “bootstrap samples”**

10

The Bootstrap Procedure

To generate a **Bootstrap Distribution** given a sample of size n from the population,

1. Generate a **bootstrap sample** of size n by resampling *with* replacement from the original sample
2. Repeat (1) a large number of times (with technology, at least 1000 times)
3. For each bootstrap sample, calculate the appropriate statistic (called the **bootstrap statistic**)
4. The collection of the bootstrap statistics form the **Bootstrap distribution**.

Q: How does this general process of generating a bootstrap distribution differ from the process of generating the *sampling* distribution?

- We sample from the original sample here; for sampling distributions, we sample from the population
- We sample with replacement here; for sampling distributions, we sample without replacement

03:00

11

Proof of Concept

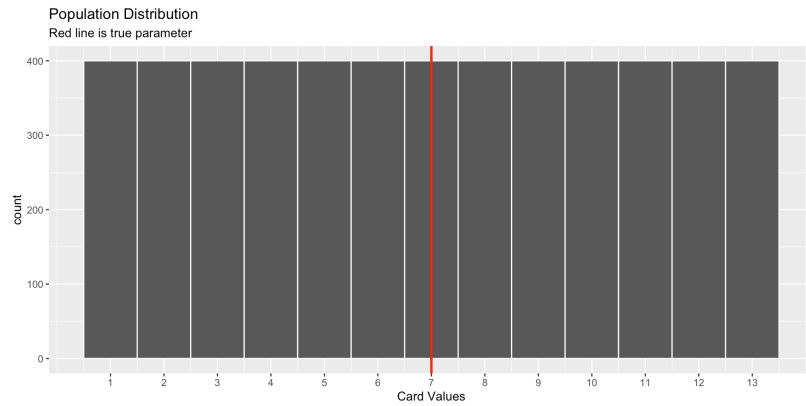
- **Population**: Consider a very large deck of cards (5200 cards) with 400 of each card value.
- **Question**: What’s the mean value from the deck of cards?
- **How We’ll Answer**:
 - Draw a sample hand of $n = 25$ cards, and calculate the mean value.
 - Estimate the sampling distribution (and therefore uncertainty) using the bootstrap distribution.

Since we have the deck of cards, we can look at:

1. The population distribution
2. The sampling distribution for sample means
3. The single sample’s distribution
4. The bootstrap distribution for sample means

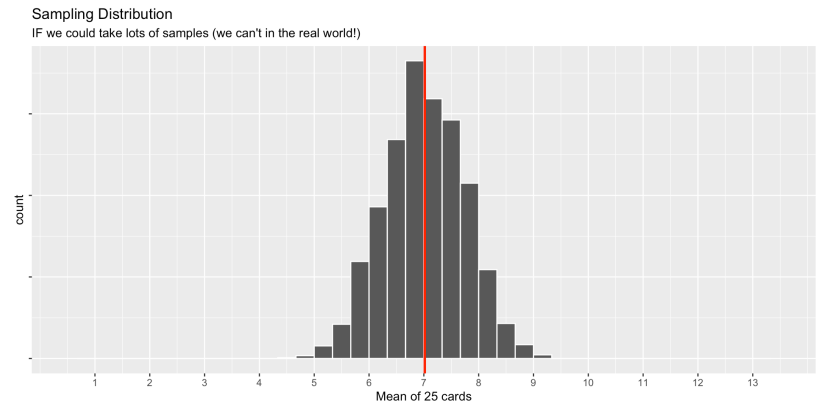
12

World's Largest Deck of Cards: Population



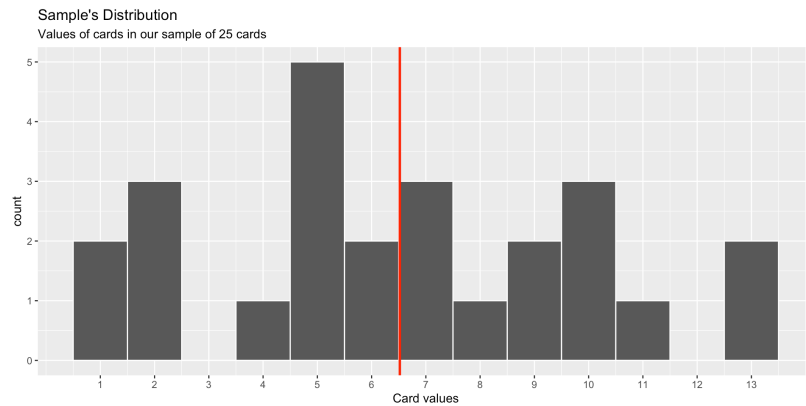
13

World's Largest Deck of Cards: Sampling Distribution



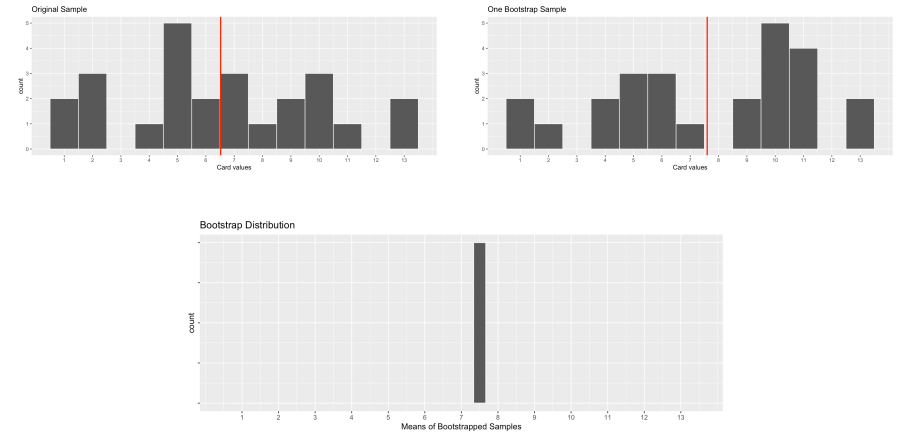
14

World's Largest Deck of Cards: Sample Distribution



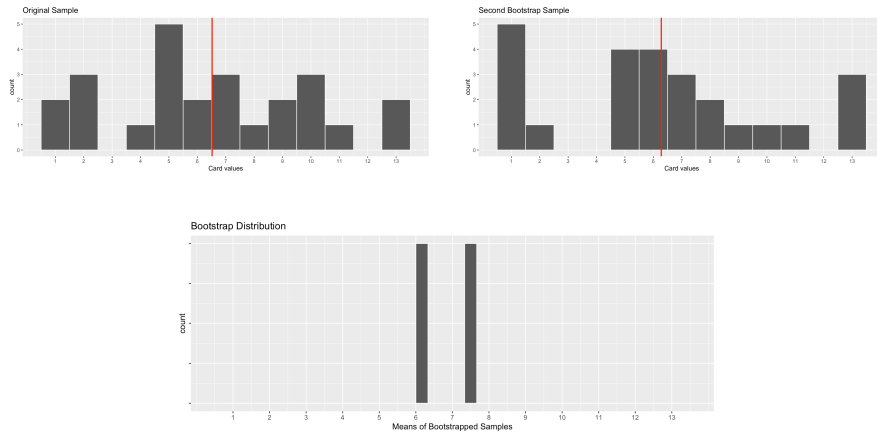
15

World's Largest Deck of Cards: Bootstrap Samples



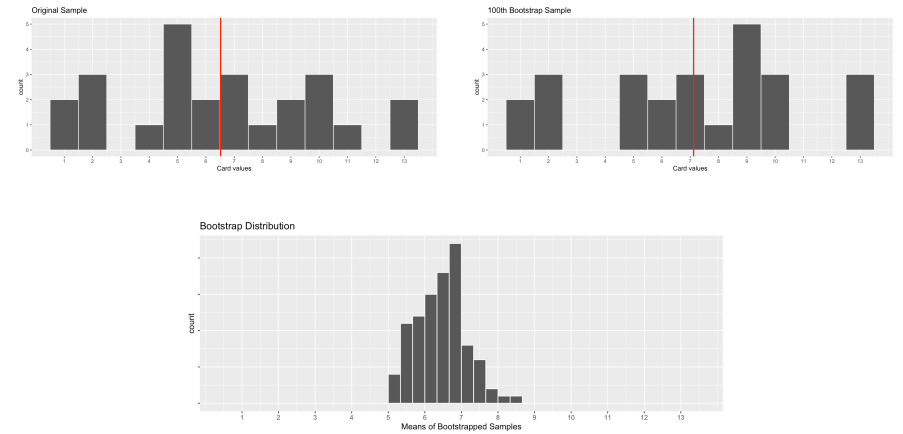
16

World's Largest Deck of Cards: Bootstrap Samples



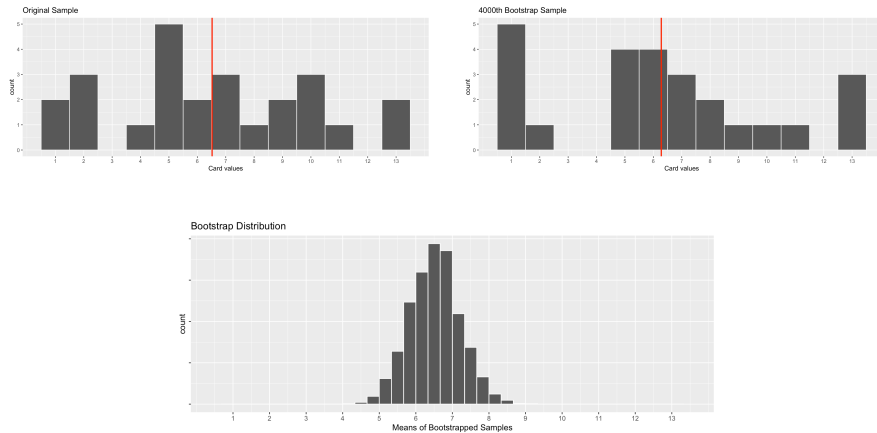
17

World's Largest Deck of Cards: Bootstrap Samples



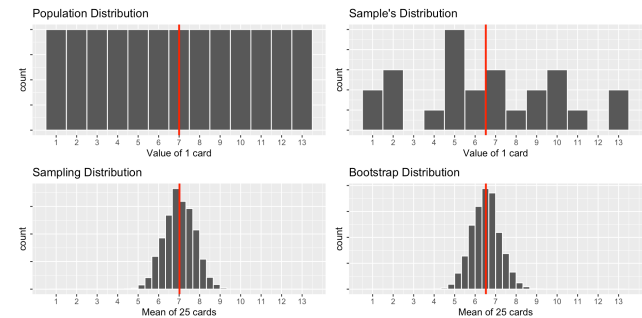
18

World's Largest Deck of Cards: Bootstrap Samples



19

World's Largest Deck of Cards: Recap



Q: Compare the Sampling Distribution and the Bootstrap Distribution. How are they similar? How do they differ?

03:00

20

World's Largest Deck of Cards

We can compute some relevant statistics:

Population:

mean_value	sd_value
7	3.742017

Sample:

mean_value	sd_value
6.52	3.513308

Sampling Distribution:

mean_xbar	sd_xbar
7.01703	0.7331422

Bootstrap Distribution:

mean_xbar	sd_xbar
6.52426	0.6902801

- Sampling and Bootstrap have slightly different means, but have similar standard deviations!
 - Q: Why are the means different?
 - Q: Does it make sense that the standard deviations are similar? Why is this helpful?
- Mean of Sampling is the *true* mean, while the Mean of Bootstrap is the *sample* mean
 - These differ because of sampling variability

02:00

21

World's Largest Deck of Cards

We can compute some relevant statistics:

Population:

mean_value	sd_value
7	3.742017

Sample:

mean_value	sd_value
6.52	3.513308

Sampling Distribution:

mean_xbar	sd_xbar
7.01703	0.7331422

Bootstrap Distribution:

mean_xbar	sd_xbar
6.52426	0.6902801

- Sampling and Bootstrap have slightly different means, but have similar standard deviations!
 - Q: Why are the means different?
 - Q: Does it make sense that the standard deviations are similar? Why is this helpful?
- The standard deviations are similar because here the sample is representative of the population, and we fix n for both distributions
 - **This is great!!** We have something very similar to SE using only our one sample!

22

Recap

- Sampling distributions are useful for understanding how close a statistic (ex. $\hat{p}$ or $\bar{x}$) might be to a parameter (ex. p or μ)
 - But we rarely know the sampling distribution. We usually only have one sample!
- Bootstrap samples allow us to **approximate the sampling distribution**.
 - Almost magically, we can use a single sample to generate many “bootstrap samples”
 - Bootstrap samples behave very much like regular samples
- Bootstrap distributions look very similar to sampling distributions,
 - ...but without the work of taking many samples!
 - The center is shifted (around the statistic instead of the parameter), but, very importantly, similar shape and **spread!**
 - The **standard deviation of the bootstrap distribution** can help us estimate the **standard error of the sampling distribution**

23